Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/03/2017
PI Name:	Uzer, Gunes Ph.D.		
Project Title:	Role of LINC Complex in Maintenance of MSC ßcatenin Signaling Under Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015 NSBRI-RFA-15-01 First Award Fellowships
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Flight Assignment:	NOTE: End date changed to 7/31/2016 (originally 10/31/2016) per NSBRI (Ed., 1/3/2017)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubin, Janet M.D. (MENTOR/ University of North Carolina)		
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	POSTDOCTORAL FELLOWSHIP The skeletal fragility that accompanies bone loss due to microgravity is a significant challenge to the success of the space mission. Mechanical signals generated during functional loading promote osteoblastogenesis at load bearing sites. Mimicking exercise, high magnitude strain (~2%), preserves the osteogenic mesenchymal stem cell (MSC) potential by regulating the active ßcatenin pool. To effectively control MSC lineage allocation, ßcatenin must be transported into nucleus, a process that is poorly understood. Our data indicates that the LINC complex (Linker of Nucleoskeleton and Cytoskeleton) is a mechanosensitive component that plays a role in the mechanical integrity of nucleoskeleton by connecting it to the cytoskeleton and participates directly in ßcatenin nuclear transport. We further showed that the loss of LINC mechanostat promotes adipogenic programs in MSC. This suggests that the unloading-induced shift to		

Task Description:	adipogenic lineage and eventual loss of functional bone tissue may be accelerated by the deterioration of the LINC mechanostat and nucleoskeletal structure. As such, strategies focused on restoring and/or improving LINC-mediated nuclear-cytoskeletal connectivity and nucleoskeletal integrity might be used to prevent microgravity related loss of MSC osteoblastic potential. Here, with the ultimate goal of identifying potential nucleoskeleton and LINC regulatory mechanisms that could be targeted to combat microgravity, the focus of this proposed fellowship will be to examine the role of nucleoskeleton and LINC complex in strain-induced (e.g., exercise) ßcatenin transport and identify the effects of simulated microgravity on the nucleoskeletal structure and connectivity as well as examining the microgravity-induced reduction in ßcatenin signaling. The following specific aims are proposed: 1) Identify the LINC elements and/or nucleoskeletal binding partners that are required for mechanically induced trafficking of ßcatenin to the nucleus; 2) Test whether microgravity induces alterations in the nucleoskeleton and components of LINC, resulting in diminished Wnt-ßcatenin signaling in MSCs. The proposed studies utilize molecular, microscopy, and proteomic techniques to study signaling and structure of primary marrow-derived MSCs. These studies will offer a new and novel examination of how cell nucleus and its connectivity to cytoskeleton as a mechanoresponsive organelle altered under microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding how mechanical signals regulate musculoskeletal tissues is the critical and fundamental first step in defining effective countermeasures and therapies. Our studies provide a first look into how the cell nucleus and its connectivity to cytoskeleton as a mechanosensitive organelle altered under microgravity and offer a potential to discover novel targets that can be targeted by both biochemical and mechanical means to mitigate microgravity or disuse induced MSC failure (decreased osteogenesis, increased adipogenesis). Our findings show that in vitro microgravity reduced MSC growth and increased nuclear height. As LINC (Linker of Nucleoskeleton and Cytoskeleton) complexes provide the connectivity between actin and nucleus, our findings suggest that microgravity induced reduction in the actin cytoskeleton may inhibit LINC functionality. We have previously shown that loss of LINC complex promotes MSC adipogenesis and limits mechanical sensitivity in MSCs. Consistent with the idea that the loss of load favors adipogenesis in MSCs, our results suggest that the reduced LINC connections that contribute to a more rounded nucleus also might explain why MSCs in microgravity are less osteogenic. Here on Earth, loss of mechanical information is a growing problem and a common theme in many conditions including but not limited to aging, obesity, and extended bedrest. Thus understanding mechanisms by which cellular structure may control the MSC fate, signaling, and ultimately musculoskeletal health is critical for improving quality of life on Earth. Our findings suggests a direct interaction between LINC complex and Wnt/ßcatenin signaling – the primary signaling axis that regulates musculoskeletal development and repair. Highlighting the importance of LINC complex mutations of the nucleoskeleton and the LINC complex lead to failure of MSC-originated tissues, causing osteoporosis, increased marrow adiposity, and musculodystrophy. Data obtained will help to better define and discover new countermeasure targets for mitigating early onset osteoporosis and other catabolic musculoskeletal changes seen where there is a lack of mechanical signals.
Task Progress:	Expanding the frontiers of spaceflight into the deep solar system requires a sophisticated understanding of microgravity-induced changes in astronaut health. A clear health concern for future space missions is that the weightless flight environment undermines bone health. The discovery of the cellular mechanisms underlying this dysfunction could enable the discovery of targeted therapies to restore the natural healing process during space travel and benefit quality of life here on Earth. The cell cytoskeleton not only modulates cell signaling and differentiation, but also -- through LINC (Linker of Nucleoskeleton and Cytoskeleton) connections -- distributes force into the nucleus thus altering gene expression by mechanisms currently unknown. We recently showed that disconnecting the nucleus from the cytoskeleton by disrupting the LINC complex limits the ability of mesenchymal stem cells (MSCs) to respond to their mechanical environment. As similar disruptions in LINC underlie muscle wasting disorders, we hypothesized that nucleocytoskeletal connectivity as facilitated by LINC would affect MSC function and fate. A LINC complex is made of multiple components. Giant Nesprin Proteins 1&2 anchored to nucleus by Sun proteins 1&2 that bind to the inner lamin A/C network. We have shown that disabling LINC function via siRNA treatment against Sun1 and Sun2 proteins (siSUN) promoted adipogenesis of MSCs. As such, the LINC complex may serve as a critical regulator of MSC fate. Our data indicate that subjecting MSCs to 1D-clinorotation, an in vitro analog to simulate microgravity, decreases MSC proliferation and diminishes actin density over the apical surface of the nucleus, resulting in increased nuclear height. These results are similar to the deletion of LINC complexes where loss of connectivity between nucleus and cytoskeleton releases the cytomechanical constrains on the nucleus leading to diminished mechano-responsivity. Our findings indicate that ßcatenin levels decrease in LINC deficient MSCs, and we have found that the deletion LINC elements reduced the basal levels of ßcatenin in MSCs. Further, LINC complex deletion decreases mechanically induced ßcatenin nuclear translocation. As reduced levels of ßcatenin are linked with MSC adipogenesis, our findings that LINC is required for maintaining basal MSC multipotentiality contributes to our knowledge in discovering why microgravity may increase bone fragility and early onset of osteoporosis during weightlessness.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
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